DAY 2 MINI PROJECT

ENTERPRISE PUBLIC VS PRIVATE IP ADDRESS ANALYSIS

📌 Project Objective

The objective of this project is to analyze public and private IP addresses, understand NAT communication, and explore how organizations use IP-based security controls in enterprise IAM environments.

This project also demonstrates practical networking analysis using real-world troubleshooting and IAM security concepts.

🌐 Introduction

Every device connected to a network requires an IP address for communication.

IP addresses are categorized into:

1. Public IP Address
2. Private IP Address

Understanding these concepts is essential in:

- Networking
- Cybersecurity
- Cloud Security
- Identity & Access Management (IAM)

🔒 Private IP Address Analysis

My Private IP Address

My private IP

10.221.64.26

Explanation

A private IP address is used inside local/internal networks.

Private IP addresses:

- Are not directly visible on the internet
- Are assigned internally by routers
- Help organizations securely manage internal communication

🌍 Private IP Ranges

10.0.0.0 – 10.255.255.255

172.16.0.0 – 172.31.255.255

192.168.0.0 – 192.168.255.255

🌐 Public IP Address Analysis

My Public IP Address

My public IP)

Example:
106.192.36.233

ISP Information

My ISP name is:
Bharti Airtel Ltd.

Explanation

A public IP address is internet-visible and assigned by the Internet Service Provider (ISP).

Public IP addresses allow communication between local networks and the internet.

🔄 NAT (Network Address Translation)

NAT converts:
Private IP → Public IP

Example:
10.221.64.26
↓
106.192.36.233

📌 Why NAT Is Important

NAT helps:

- Save IPv4 addresses
- Improve security
- Hide internal devices
- Allow internet communication

🌐 Network Communication Flow

Laptop
↓

Private IP
↓
Router/Gateway
↓
NAT Translation
↓
Public IP
↓
ISP
↓
Internet

🗌 DNS Analysis

DNS (Domain Name System) converts domain names into IP addresses.

Example:
google.com → IP address

This helps users access websites using names instead of numerical IP addresses.

🛠 Commands Used

ipconfig

ping google.com

nslookup google.com

📌 Ping Analysis

The ping command was used to:

- Test internet connectivity
- Measure response time
- Verify communication with external servers

📌 DNS Lookup Analysis

The nslookup command was used to:

- Analyze DNS resolution
- Convert domain names into IP addresses

🔐 IAM Security Mapping

Organizations use IP addresses in IAM systems for:

- Conditional Access

- Geo-location tracking
- MFA triggering
- Suspicious login detection
- VPN restrictions
- Risk-based authentication

🎯 Enterprise Scenario Analysis

Scenario

An employee usually logs in from India.

Suddenly, a login attempt occurs from another country using an unknown public IP address.

IAM Security Response

IAM systems may:

- Trigger Multi-Factor Authentication (MFA)
- Block the login
- Alert security teams
- Mark login as risky
- Apply conditional access policies

🔥 Zero Trust Security Concept

Modern organizations follow:
"Never Trust, Always Verify"

Even if login comes from a valid IP:

- Identity verification is still required
- Device trust is validated
- Risk analysis is performed

🛡 Firewall & Security Controls

Organizations use firewalls to:

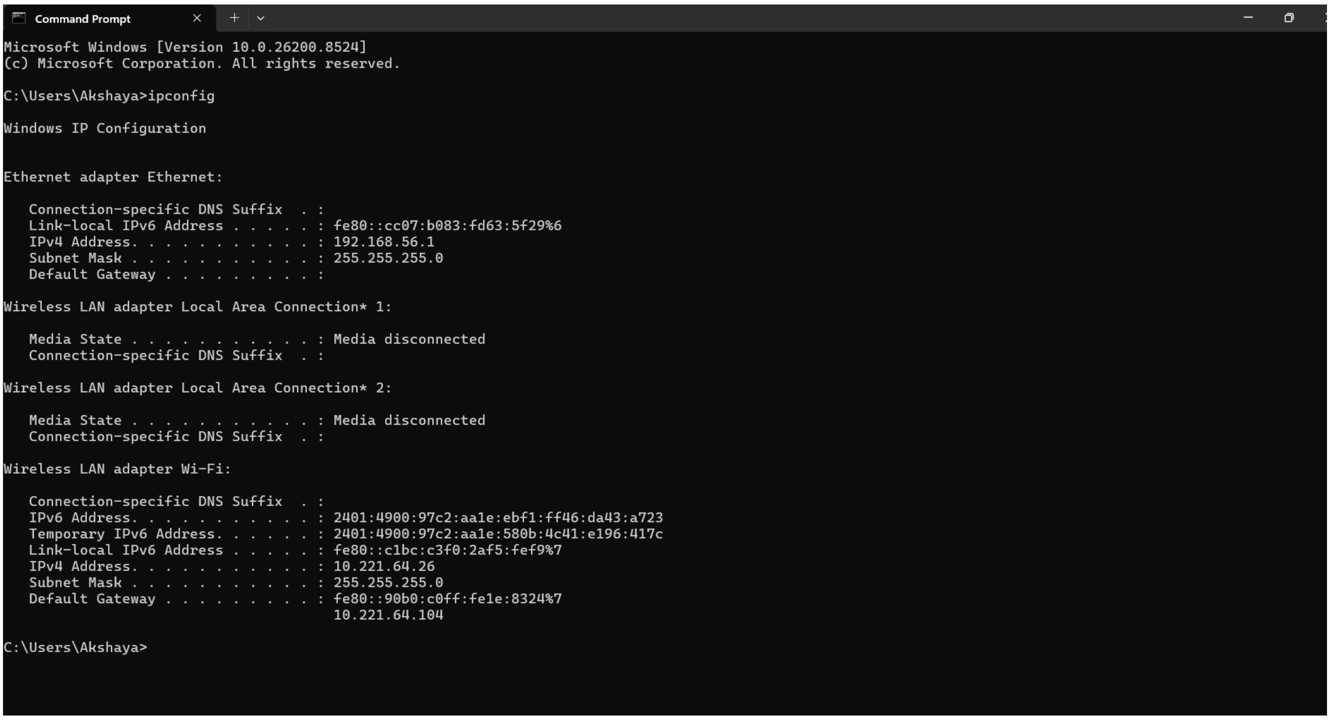
- Allow trusted IP addresses
- Block suspicious IP addresses
- Restrict traffic from risky countries
- Monitor network communication

📊 Public IP vs Private IP Comparison

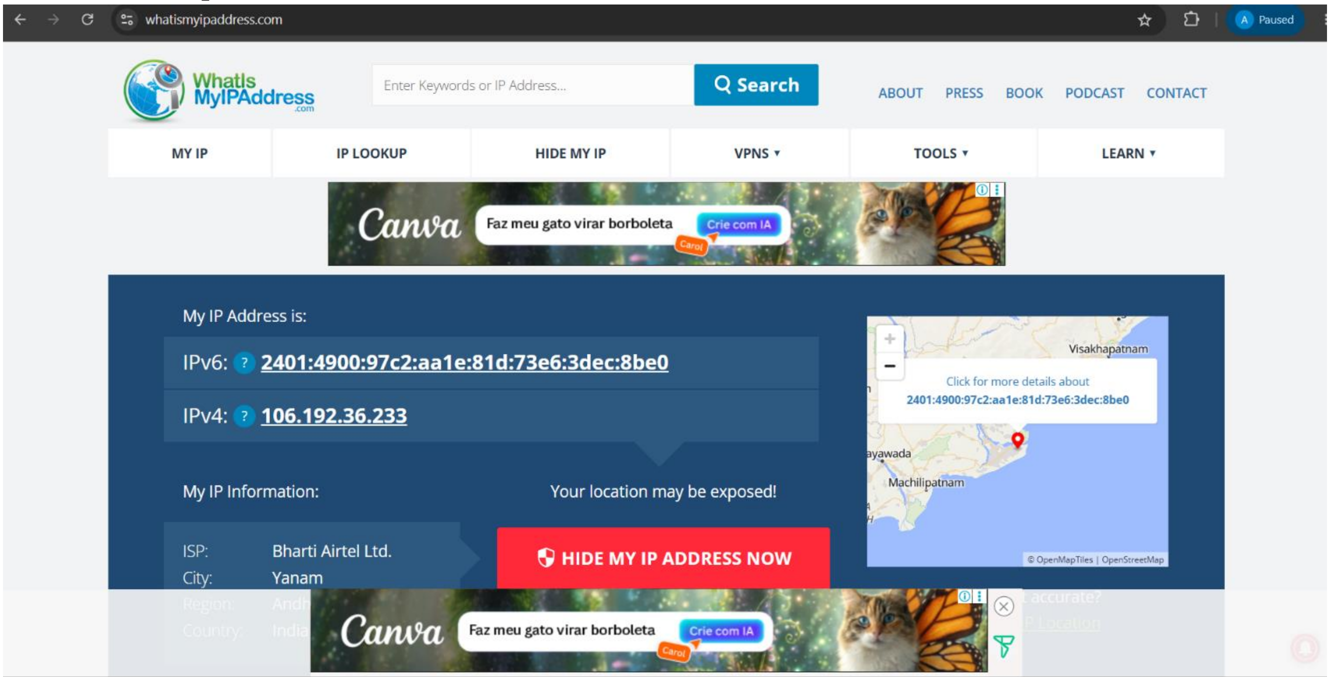
Public IP	**Private IP**
Visible on internet	Used inside local network
Assigned by ISP	Assigned by router
Globally unique	Internal use
Used for internet communication	Used for internal communication

📸 Screenshots Included

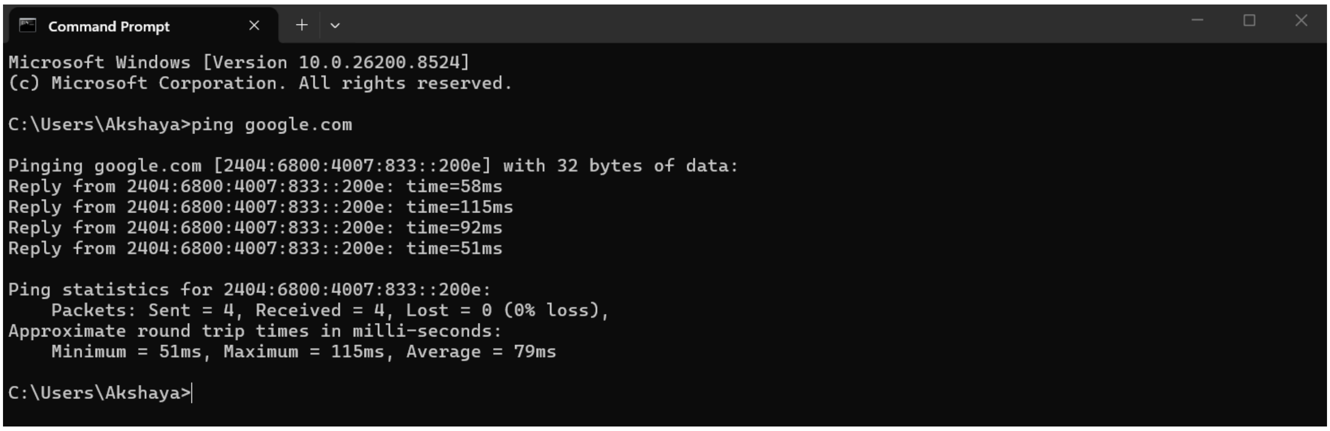
1. Private IP (ipconfig)



2. Public IP lookup

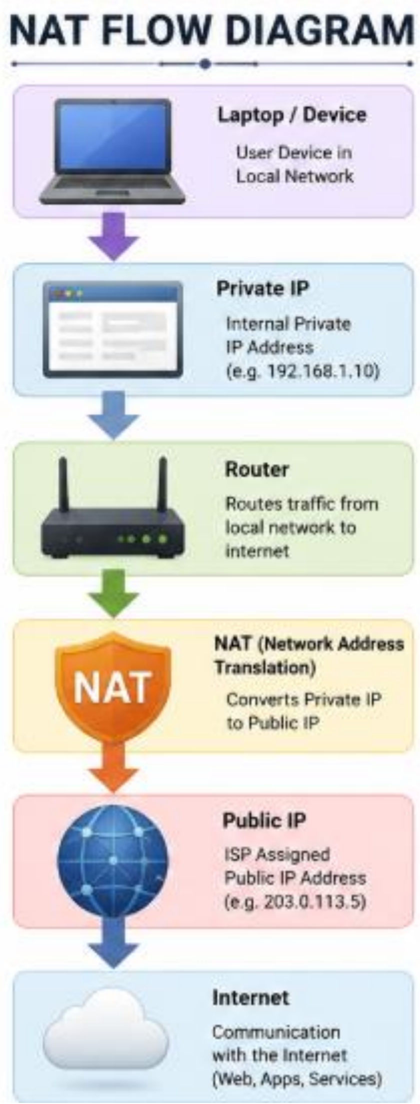


3. Ping command output



4. DNS lookup output

5. NAT flow/network diagram



🏆 Key Learning Outcome

Through this project, I learned:

- Difference between public and private IP addresses
- NAT communication flow
- DNS resolution process
- Enterprise networking concepts
- IAM security mapping using IP addresses
- Conditional access and geo-location security concepts

This project strengthened my foundational skills in:

- Networking
- Cybersecurity
- IAM
- Cloud Security
- Security Operations